FRIDAY CHALLENGE

COFFEE SALES INSIGHTS DASHBOARD

Step-by-Step Excel Guide

Objective

Create an interactive Coffee Sales Insights Dashboard using the provided coffee vending machine dataset. The dashboard should show which coffees sell best, how customers pay, and when sales peak, and present the findings through KPIs, charts, slicers, and key insights.

About the Dataset

Every row is one cup sold by a coffee vending machine between 1 March 2024 and 23 March 2025 (3,636 sales). Amounts are in the machine's local currency, so report them as plain numbers.

Column	What it means
date	Day of the sale
datetime	Exact date and time of the sale
cash_type	How the customer paid: card or cash
card	Anonymous card ID (blank for cash payments, this is expected)
money	Price paid for the cup
coffee_name	Drink sold (8 types, e.g. Latte, Americano, Cappuccino)

Source: Coffee Sales dataset by Yaroslav Isaienkov on Kaggle (kaggle.com/datasets/ihelon/coffee-sales).

STEP 1 : Prepare the Data

The PivotTables need a Month and an Hour for every sale, so add two helper columns first.

1. Click any cell in the data and press Ctrl + T to turn it into an Excel Table.
2. In column G, type the header **Month** and the formula **=TEXT(A2,"yyyy-mm")**
3. In column H, type the header **Hour** and the formula **=HOUR(B2)**
4. The Table fills both formulas down automatically. Check that the last row (row 3637) has a Month and an Hour.

STEP 2 : Create the PivotTables and Charts

Insert each PivotTable on its own sheet (Insert > PivotTable > New Worksheet) and then add a PivotChart (PivotTable Analyze > PivotChart).

5. **Revenue by Coffee** : Rows = coffee_name, Values = Sum of money. Sort largest to smallest. Chart: Clustered Bar.
6. **Revenue by Payment Type** : Rows = cash_type, Values = Sum of money. Chart: Pie.
7. **Revenue by Month** : Rows = Month, Values = Sum of money. Chart: Line.
8. **Cups Sold by Hour** : Rows = Hour, Values = Count of coffee_name. Chart: Clustered Column.

Important: Revenue is the SUM of money. Cups sold is the COUNT of rows. If a Values field says "Count of money" where you need revenue, click it, choose Value Field Settings, and change it to Sum.

STEP 3 : Convert Payment Revenue into Percentages

For the payment chart, show each payment type as a share of total revenue.

9. Right-click a value in the Payment Type PivotTable.
10. Select Show Values As.
11. Select % of Grand Total.
12. Format the pie chart data labels to show the category name and percentage.

Example: If card shows 97%, it means 97% of all revenue was paid by card. It does not mean 97% of customers prefer cards.

STEP 4 : Create the KPI Cards

Create four KPI cards at the top of the dashboard. Build them from a small KPI PivotTable so they also react to the slicers.

13. Insert a new PivotTable on a sheet called KPI. Leave Rows empty and add three Values: Sum of money, Count of coffee_name, and Average of money.
14. On the dashboard, draw a shape for each card, click the shape, type = in the formula bar, and click the matching KPI cell. Excel writes a GETPIVOTDATA formula for you.

KPI	Calculation
Total Revenue	Sum of money.
Cups Sold	Count of coffee_name (one row = one cup).
Avg. Price per Cup	Average of money (same as Total Revenue ÷ Cups Sold).
Card Share	The card value from the Payment Type PivotTable (already % of Grand Total). Wrap it in IFERROR(...,0) so it shows 0% when the Cash slicer is selected.

STEP 5 : Add Slicers

15. Click any PivotTable and insert slicers for coffee_name, cash_type, and Month.
16. Connect each slicer to all the PivotTables using Slicer > Report Connections (or PivotTable Connections).
17. Test each slicer: select Latte, then Cash, and check that every KPI card and chart updates. Clear the filters when done.

STEP 6 : Format the Dashboard

STEP 7 : Generate Key Insights

Write three short insights based on the final dashboard (with all slicers cleared). Describe what the charts show, and remember that March 2025 is a partial month (data ends on 23 March).

Final Checklist

- ☐ Month and Hour helper columns added.
- ☐ Four PivotTables and charts created with the correct fields.
- ☐ Revenue uses Sum of money; cups use Count.
- ☐ Payment chart shown as % of Grand Total.
- ☐ 4 KPI cards completed.
- ☐ Coffee, Payment Type, and Month slicers added and connected to all Pivot Tables.
- ☐ Dashboard formatting is clean and consistent.
- ☐ Key insights added.

REFERENCE DASHBOARD

Use the dashboard below as a visual reference for the expected layout, formatting, charts, slicers, and insights. Your dashboard does not have to be identical, but it should contain the required components.

